Copernicus Global Land operations
”Vegetation and Energy”
“CGLOPS-1”

Framework Service Contract N° 945120-IPR-2023

ALGORITHM THEORETHICAL BASIS DOCUMENT

SOIL WATER INDEX (SWI) 0.1°
SOIL WATER INDEX 10 DAYS (SWI10) 0.1°
SWI TIME SERIES (SWI_TS) 0.1°

Issue I1.40

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List of Acronyms

AD	Applicable document
ASCAT	Advanced Scatterometer (Metop)
ATBD	Algorithm Theoretical Basis Document
BUFR	Binary Universal Form for the Representation of meteorological data
CAL/VAL	Calibration and Validation Group of CEOS
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
COG	Cloud-Optimized Geotiff
DGG	Discrete Global Grid
EAPG	EUMETSAT ASCAT Product Guide
ECMWF	European Centre for Medium Range Weather Forecasting
ECV	Essential Climate Variable
EPS	EUMETSAT Polar System
ERA-40	ECMWF ReAnalysis 40 data set
ERA-Interim	ECMWf Reanalysis Interim
ERS	European Remote Sensing Satellite
EU	European Union
EUMETCast	A service run by EUMETSAT for data dissemination
EUMETSAT	European Organization for the Exploitation of Meteorological Satellites
GCM	Global Climate Model
GEM6	Generation of weather Elements for Multiple applications
GBOV	Ground-Based Observations for Validation
GIO	GMES Initial Operations
GLWD	Global Lakes and Wetlands Database
GMES	Global Monitoring for Environment and Security
HDF	Hierarchical Data Format
METOP	Meteorological Operational satellite
NETCDF	Network Common Data Form
NRT	Near real time
PUM	Product User Manual
QAR	Quality Assessment Report
QFLAG	Quality Flag
SSF	Surface State Flag

SSM	Surface Soil Moisture
SSM/I	Special Sensor Microwave Imager
SWI	Soil Water Index
UMARF	Unified Meteorological Archive and Retrieval Facility
WGS84	World Geodetic System 1984

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

The CLMS portfolio includes a global Soil Water Index (SWI) at 0.1° spatial resolution derived from the observations from the Advanced Scatterometer (ASCAT) on board the Meteorological Operational (Metop) satellite.

The document provides a concise review of the background, application and development of the Soil Water Index (SWI). The SWI calculated over a 10-days time period (SWI10) is described as a post-processing. The SWI product is also re-formatted to be delivered as historic time series (SWI_TS).

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The document contains the theoretical basis of the SWI methodology and introduces a method for determination of soil surface state, which provides further improvement for SWI by flagging freeze/thaw conditions. It presents also the time compositing module applied as post-processing to calculate the 10-days SWI (SWI10). It describes as well as the re-formatting of historic products to make easier the handling of products by users working at local and regional scales, interested in obtaining harmonized historic SWI time series (SWI-TS) processed with the latest algorithms of both surface soil moistures and SWI.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 introduces the requirements
- Chapter 3 describes the methodology in detail. It provides the basic underlying assumptions, related and previous applications, a description of the necessary input data as well as information on product accuracy and its limitations.
- Chapter 4 presents the post-processing: calculation of SWI10 and re-formatting of historic products as time series (SWI_TS)

1.3 RELATED DOCUMENTS

1.3.1 Applicable document

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Output

Document ID	Descriptor
CGLOPS1_PUM_SWI0.1	Product User Manual of the Version 4 of the global Soil Water Index product at 0.1° resolution
GIOGL1_QAR_SWIV3-SWI10	Validation Report describing the results of the scientific quality assessment of the SWI V3.2 and SWI10 product

These documents are available on https://land.copernicus.eu/en/technical-library

1.3.3 External documents

EPS Product File Naming for EUMETCast:

https://www-cdn.eumetsat.int/files/2020-04/pdf_eps_prod_naming.pdf

Eumetsat ASCAT Products Guide:

https://user.eumetsat.int/s3/eup-strapi-media/ASCAT_User_Guide_1618a3d741.pdf

ASCAT_SSM_NRT_12_5km_H29_PUM_v0_2: ASCAT SSM H29 Product User Manual available at: https://hsaf.meteoam.it/Products/Detail?prod=H29

ASCAT_SSM_NRT_12_5km_H29_PVR_v0_1: ASCAT SSM H29 Product Validation Report available at: https://hsaf.meteoam.it/Products/Detail?prod=H29

2 REQUIREMENTS

According to the applicable document [AD1], the requirements relevant for SWI are described below.

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

SWI – Soil Water Index	
The surface soil water index provides daily information about moisture conditions in different soil depths. Soil moisture is a very heterogeneous variable and varies on small scales with soil properties and drainage patterns. Satellite measurements integrate over relative large-scale areas, with the presence of vegetation adding complexity to the interpretation. The SWI quantifies the amount of water (m^3/m^3) in soil layers at various depths and is derived from the SSM using a water balance model. The freeze and thaw status of soil is provided.	
Geometric Properties:	
spatial resolutions	3-10km/1km
geolocation precision	Better than 0.5 pixels
coordinate position	Centre of the pixel
geodetical datum	WGS84
geographic projection	regular latitude/longitude grid
Geographic Coverage:	Global/EU
Time span of output product:	SWI: Daily SWI10: 10-day period SWI TS (timeseries): 6 months
Timeliness of output delivery:	SWI: 12hrs SWI-10: within 3 days after synthesis period
Uncertainty	Threshold: 0.1 m^3/m^3 Goal: 0.04 m^3/m^3
Stability	Threshold: 5% Goal: 3%

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the CA during the FWC.

2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS-245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Level 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

Land Surface Temperature (LST) and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.
The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 METHODOLOGY DESCRIPTION

3.1 OVERVIEW

Active microwave remote sensing of soil surfaces in C-band usually gives information about the first few centimeters of a soil profile. Besides information on the status of the surface state and soil moisture conditions also information on the moisture conditions within the underlying soil profile is of interest for different applications.

3.2 THE RETRIEVAL ALGORITHM

3.2.1 Soil Water Index

3.2.1.1 Basic underlying assumptions

The moisture contained within a soil profile is mainly derived from precipitation via the process of infiltration. The Soil Water Index (SWI) algorithm is based on a simple model describing this infiltration process. As input temporally irregular surface soil moisture (SSM) measurements are used. The SSM is retrieved from scatterometer observations from the ASCAT instruments on board the MetOP satellites according to the algorithm described in (Wagner 1998). These SSM observations represent the moisture content in a thin top soil layer (1-5 cm). The infiltration process is then modelled according to Wagner et al. (1999) with *SWI* being calculated for time t_n integrating *SSM* data over a preceding time period *T* using the formula:

$$SWI(t_n) = \frac{\sum_{i}^{n} SSM(t_i) e^{-\frac{t_n - t_i}{T}}}{\sum_{i}^{n} e^{-\frac{t_n - t_i}{T}}} \quad for\, t_i \le t_n \qquad \text{(Eq. 1)}$$

In which t_n is the observation time of the current measurement and t_i are the observation times of the previous measurements. These times are given in Julian days. In (Eq. 1) all SSM observations made before t_n are summed up and exponentially weighted. The factor T determines how fast the weights become smaller and how strongly SSM observations taken in the past influence the current SWI. If e.g. the T value is 5 and a SSM measurement was taken 10 days before t_n then this observation would receive the weight $e^{-\frac{10}{5}} = 0.135$. Using a T value of 20 in the same case would increase the weight to $e^{-\frac{10}{20}} = 0.607$.

This formulation represents a simple two-layer water balance model, with the first layer as the soil surface layer accessible to C-band scatterometers and the second layer as the part of the profile which extends downwards from the bottom of the soil surface layer. The second layer is assumed to serve as a reservoir which is connected to the atmosphere via the first layer. The moisture conditions in the first layer, which is in direct contact with the atmosphere, is temporally highly dynamic. With increasing profile depth in the second layer, the temporal dynamic of the moisture conditions is decreasing, as the amount of water stored in the reservoir depends on the infiltration of water added to the first layer during precipitation events. Therefore, the water content of the reservoir layer is solely controlled by the past moisture conditions in the surface layer and thus the precipitation history. Here, more recent precipitation events have a stronger impact on the moisture conditions in the reservoir, which is accounted for by the exponential weighting function in Eq. 1. The link between the remotely sensed surface layer Θ_s and the reservoir Θ is described by (Ceballos et al. 2005):

$$L\frac{d\Theta(t)}{dt} = C\cdot\left[\Theta_s(t) - \Theta(t)\right], \qquad \text{(Eq. 2)}$$

where t is time, L the depth of the reservoir layer and C an area-representative pseudo-diffusivity constant. If C is assumed to be constant and $T=L/C$, then the differential equation (Eq. 2) can be solved as follows:

$$\Theta(t) = \frac{1}{T}\int_{-\infty}^{t}\Theta_s(t')\exp\left[-\frac{t-t'}{T}\right]dt' \qquad \text{(Eq. 3)}$$

The output data range is the same as the input data range which for ASCAT SSM i.e. 0 - 100%.

3.2.1.2 Model Limitations

Most importantly evapotranspiration is not considered although it is of great importance to the soil water balance. Studies have investigated approaches for deriving root zone soil moisture through more physically based models. (Manfreda et al. 2014) introduced a two layer model that, among other parameters, considers soil porosity, relative saturation at the wilting point, layer depth and a soil water loss coefficient (V). V was parameterized to account for both, evapotranspiration and percolation losses. In some cases the validation of this model against in situ stations showed improvements when compared to the SWI method. Since this model relied on station specific soil data it is at the moment not clear how and if it could be applied globally.

It is also assumed that the soil hydraulic conductivity is constant while in reality it may vary over orders of magnitude depending on soil moisture conditions (Hillel 1982).

3.2.1.3 Related and previous applications

The performance of SWI data derived from ERS-1/2 C-band scatterometer data has been validated by different researchers and has been published in a number of publications. In a study comparing model-derived soil moisture and scatterometer-derived SWI information Parajka et al. (2006) found good agreements between the two data sets especially for the lower parts of Austria. For a catchment in Upper Austria, a correlation of r=0.78 has been calculated. Alpine areas are found to be problematic due to complex terrain and frozen/snow covered soil conditions for extended periods of time. Additionally, the potential of SWI data for improving the performance of hydrological models in both gauged and ungauged basins has been examined. It could be shown that the assimilation of scatterometer-derived SWI in the calibration phase can improve the relationship between the two soil moisture data sets substantially for gauged basins. A weaker correlation was only detected in Alpine areas, where the scatterometer derived SWI is affected by the extreme topography and by snow cover and/or frozen soil conditions.

Zhao et al. (2008) used monthly averages of ERS scatterometer derived *SWI* for the period 1992-2000 to examine water availability and precipitation distribution in China. The remotely sensed SWI is compared to monthly averaged relative *in-situ* measurements of soil moisture from Chinese meteorological stations as well as to monthly averages of precipitation derived from the Global Precipitation Climatology Center at three different scales – local, regional and statewide. The analysis showed, that the scatterometer derived *SWI* is a good descriptor of the water availability at all three scales. It has also been shown, that the *SWI* can detect extreme events such as floods and droughts.

A study by de Wit & van Diepen (2007) investigates the assimilation of ERS scatterometer derived *SWI* data into a water balance model of the world food studies crop model using an Ensemble Kalman filter for different European test site for the years 1992-2000. Results were evaluated using existing crop yield statistics. The study showed that the assimilation of SWI data into the crop model could improve the model results. Best results were achieved for winter wheat and grain maize.

Modeled soil moisture for a test site in Southern France was compared to *SWI* data from the ERS scatterometer by Pellarin et al. (2006). It could be shown, that in general a good agreement between the modeled and the remotely sensed soil moisture products could be achieved. Sensitivity studies showed that the agreement depends on the temporal sampling characteristics, the errors connected to surface soil moisture *SSM* and the consideration of frozen soil conditions.

Soil moisture extracted from thermal spectral data from the Meteosat satellite was compared to ERS *SWI* data and in-situ soil moisture measurements in a study by Verstraeten et al. (2006). Test statistics

have been prepared showing correlation coefficients up to r=0.91. Further data analysis showed, that choosing different values for T has strong impact on the *SWI* and the comparison to other soil moisture data sets. They propose to use climate zone specific selection of T instead of using a fixed T value.

Dirmeyer et al. (2004) have compared the SWI with seven other global wetness products, three produced by land surface model calculations, three from coupled land atmosphere reanalysis, and the so-called Soil Wetness Index data set derived from SSM/I data (Basist et al. 1998). They found that, while the SSM/I data clearly have a different character from all the other data sets, the ERS scatterometer data revealed many similarities with the modeled wetness products

.

3.2.1.4 Alternative methodologies currently in use

Alternative approaches for estimating the SWI comprise hydrologic modeling including the application of soil type specific infiltration models or use of assimilation schemes combining data of various sources such as in-situ observations and remote sensing data. The parameterization of hydrologic models usually requires a large amount of input data and a thorough knowledge of the region, for which the hydrologic model is implemented. With increasing size of the region, this task becomes more demanding. Data assimilation at continental to global scales requires appropriate models and is computationally expensive. Although these methods provide an alternative approach for deriving profile soil moisture, the SWI algorithm is comparably simple and only dependent on the availability of ASCAT scatterometer derived surface soil moisture data as input parameter.

3.2.2 Surface State Flag

3.2.2.1 Basic underlying assumptions

Soil moisture retrieval from scatterometer data has certain limitations. Soil moisture cannot be retrieved when the surface is frozen or covered by snow, dense vegetation, or water. While dense vegetation and water on the surface are almost static factors and detectable from backscattered signal, it is not as straight forward from the scatterometer measurements themselves to say if the ground is frozen or snow covered. Backscatter measurements from densely vegetated areas indicate very stable signal with low sensitivity due to strong volume scattering. Contrary to vegetated surface, the interaction of scatterometer signal with water is more or less specular and therefore the backscatter intensity is very low and results in very high measurement noise. These two limiting factors can be flagged using surface soil moisture noise information which is attached to the SWI data structure. The snow and frozen surface conditions are dynamic factors and should be flagged carefully to avoid miscalculation of SWI. If such ground conditions are not detected and masked then the calculated SWI is seriously affected. For example, Pellarin et al. (2006) showed over a region in south-western

France that short term frost events caused artificially low values in the ASCAT surface soil moisture data. Without masking, the SWI was thus too low, while with proper masking it reflected profile soil moisture measurements quite well.

Several studies have investigated the applicability of scatterometer data in various cryosphere research areas; for instance, mapping snowmelt extent (Wismann et al., 1997; Wismann, 2000), snow accumulation in Greenland (Drinkwater et al., 2001), snow cover over the Northern Hemisphere (Nghiem et al., 2001), frozen terrain in Alaska (Kimball et al., 2001). Other studies have used scatterometer data for determination of freeze/thaw cycles in Northern Latitudes (Bartsch et al., 2007), spatial and temporal variability of sea ice (Drinkwater et al., 2000), classification of sea ice in Polar Regions (Remund et al., 2000), deriving the surface wind-induced patterns over Antarctica by measuring the azimuthal modulation of backscatter (Long et al., 2000).

In SWI product versions before 4.0, a numerical algorithm developed at TU Wien was used for detection of freeze/thaw conditions based on the parameters and thresholds extracted by processing of the ASCAT normalized backscatter time series and using distinctive decision trees. In this method a Surface State Flag (SSF) is attached to each backscatter measurement which shows frozen/unfrozen surface, thaw condition/snow melt (Table 1).

In product version 4.0, meteorological forecast from ECMWF became available, and the SSF flag has been updated to use this data. ECMWF provides forecasts for skin temperature and snow depth every 2 hours. Similar to the advisory flag of the H29 Metop ASCAT Surface Soil Moisture Near Real Time 12.5 km product[3], we derive binary frozen soil and snow cover masks from forecasts of skin temperature and snow depth. For each forecast within the 24-hour time-frame of the SWI product, we create 12 such binary masks where skin temperature is below 0° C or snow depth is above 0 m. Then the SSF flag is set to “Frozen” for a certain point if the majority of these 12 masks indicate frozen surface conditions at that point. If there is no forecast available the flag is set to ”Unkown”. Over areas where no meteorological forecasts for land surface variables are available, such as the ocean, the flag is set to “Could not be determined“. The remaining flag “Temporary water on the surface” is deprecated, but is still kept in the data format to allow for backwards compatibility.

In this new product version 4.0, the SSF mask is not used anymore to mask ASCAT surface soil moisture data, since this is now done utilizing the advisory flags of the new H29 surface soil moisture product. Concretely, we mask each SSM observation where the advisory flags indicate frozen soil, see Section 3.2.3.2 for more details. The SSF mask, however, is kept as an auxiliary data for the users.

[3] https://hsaf.meteoam.it/Products/Detail?prod=H29

SSF value	Detected surface state
255	Could not be determined
0	Unknown
1	Unfrozen
2	Frozen
3	Temporary water on the surface

Table 1: Possible values of the Surface State Flag

3.2.2.2 Model limitations

Radar backscatter from very dry surfaces can be very similar to backscatter from flooded areas. Because of that, the possibility exists that these very dry events are wrongfully flagged by the SSF algorithm.

Furthermore, in product version 4.0, the meteorological forecast data might not be available in NRT (during the daily processing) to calculate the SSF flag. In such cases the flag is set to "Unknown". The backscatter-based SSF algorithm is not used anymore in product version 4.0, hence the following limitations of the backscatter based SSF algorithm is only applicable for prior versions. The source of the water on the surface cannot be established without doubt. Since the backscatter based SSF algorithm uses the SSF "state 3" (Table 1) mainly in spring or autumn it is likely, but not certain, that the reason is snow melt most of the time.

3.2.2.3 Freeze/thaw parameters database

The magnitude of the backscattered scatterometer signal is dependent on the geometric and dielectric properties of the surface. When the soil surface freezes, dielectric properties of the soil changes significantly which usually results in low backscatter values. As snow begins to fall and accumulates over the surface, due to volume scattering, backscatter signals may increase depending on microwave frequency. The response of dry snow volume to microwaves is rather complex and depends on snow properties like snow depth, density, and average grain size as well as the age of snowpack. With increasing temperature in spring, snow begins to melt and water covers the surface of snow pack which causes a sudden drop in backscatter. After the snow melting period, soil and vegetation begin to thaw and consequently backscatter arise again. Figure 1 shows a typical example of freeze/thaw process as described above observable in ASCAT normalized backscatter at 40°.

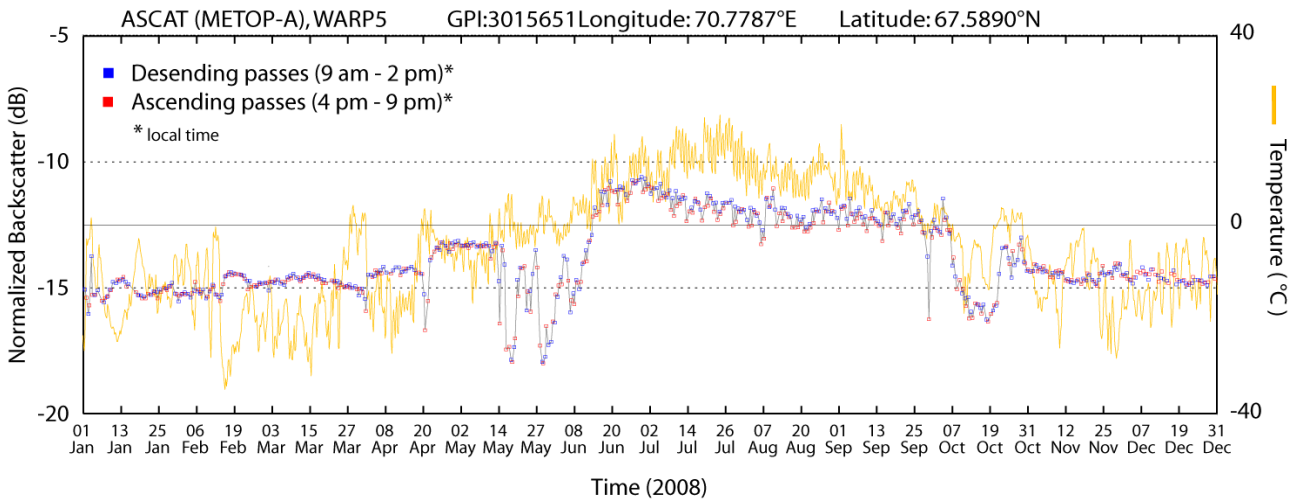


Figure 1: ASCAT normalizes backscatter at 40° compared with temperature

In TU Wien method, soil surface state is determined by evaluating the normalized backscatter measurement at 40° and using following set of parameters:

- *Mean of normalized backscatter in summer:* Mean of normalized backscatter during warm season. (June, July, August in Northern Hemisphere and December, January, February in Southern Hemisphere).
- *Mean of normalized backscatter in winter:* Mean of normalized backscatter during cold season. (December, January, February in Northern Hemisphere and June, July, August in Southern Hemisphere).
- *The first seasonal transition time*: Day of year of the first transition time (winter to summer in Northern Hemisphere)
- *The second seasonal transition time*: Day of year of the second transition time (summer to winter in Northern Hemisphere).
- *Backscatter level during frozen conditions*: Normalized backscatter value when the surface is frozen.
- *Backscatter level during snow melt (thaw condition)*: Normalized backscatter value when the surface, which is covered by snow, begins to thaw.
- *Slope of backscatter in minus temperature*: Rate of backscatter change with respect to temperature when the temperature is below zero.
- *Noise of normalized backscatter during frozen conditions*: Standard deviation of backscatter measurements below the frozen level during winter.

- *Ice flag*: An advisory flag which indicates frozen Lake or permanent ice conditions.

These parameters have been extracted for each grid point by processing of two years (2007-2008) ASCAT normalized backscatter data and comparing with modeled air temperature data from the ECMWF ReAnalysis (ERA-Interim) dataset. The ERA-Interim dataset is based on the ECMWF Integrated Forecast Model, a global numerical weather prediction model.

Figure 2 indicates normalized backscatter values during frozen conditions. Colors represent backscatter thresholds at frozen level in global scale and white areas indicate unknown conditions where either the method failed to estimate the frozen level or there were not enough backscatter measurements at minus temperature to calculate the frozen level.

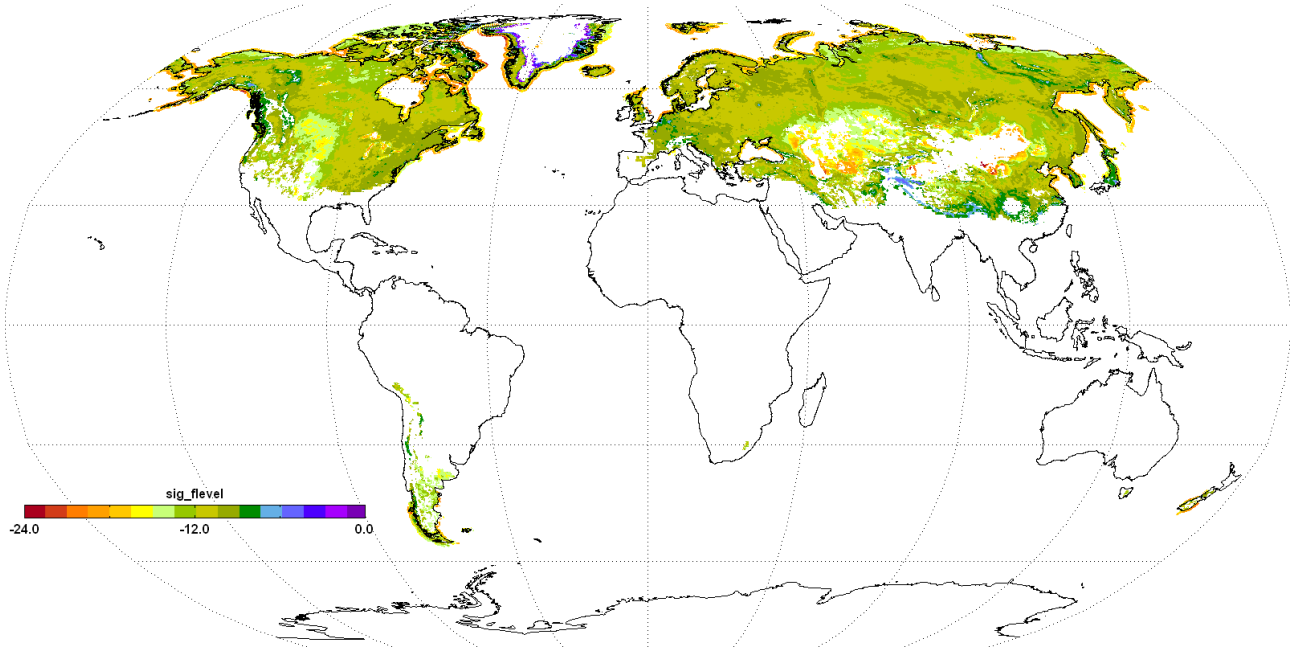


Figure 2: Frozen level of normalized backscatter in global scale which has been retrieved by processing of 2 years (2007-2008) ASCAT level-1 data (from WARP5.2 database)

3.2.2.4 Eliminating backscatter outliers using climatological quality flags

Backscatter measurements mostly show a linear gradient around 0°C but do not behave in easily describable patterns at high or low temperatures. It can occur that backscatter has the same value at +20°C and -20°C, these measurements cannot be differentiated using the radar backscatter alone. Because of this, the climatological quality flags that are provided with the ASCAT input data are used to correct for very unlikely SSF. The quality flags provide probabilities for frozen ground, snow and ice cover, for every DGG point, for every day of the year, ranging from 0 to 100%. The underlying

assumption is that if any of the probabilities is more than 95% then the SSF is set to frozen and if the probabilities are below 5% then the SSF is set to unfrozen. More information about ASCAT advisory flags can be found in Scipal et al. (2005).

3.2.3 Methodology

In SWI product versions before 4.0, the ASCAT Level2 products are delivered in BUFR format with arbitrary pixel locations via EUMETCast and are used as input for the calculation of *SWI*. In the first step, the *SSM* input data have to be resampled to a Discrete Global Grid (DGG). The DGG is an adapted Sinusoidal grid generated using the GEM6 model based on an ellipsoid. The grid is defined such that the spacing is approximately 12.5 km. In total, the grid consists of 3,264,391 points, where 839,826 points are over land. The underlying satellite data are sampled to the DGG grid. For more information about the grid definition see also Kidd (2005)[4]. For resampling a weighting function (Hamming window) is used which is implemented based on the distance between the grid point and the location of the measurement.

In SWI product version 4.0, the input data stream has been upgraded to use the new H29 ASCAT SSM product. This input product is already delivered on a matching fixed Earth grid sampled at 12.5 km. Hence, this resampling step is skipped in the new version 4.0.1 and the data is used directly without input resampling.

3.2.3.1 Recursive calculation of SWI

The calculation of the *SWI* values using the approach introduced by Wagner et al. (1999) requires the availability of historic *SSM* time series data. A computational adaption of this *SWI* algorithm has been proposed by Albergel (2008). Using a recursive reformulation of the *SWI*, the calculation can be started at any time. The *SWI* at time *t* for a given characteristic time length *T* can be derived using the formula:

$$SWI_T(t_n) = SWI_T(t_{n-1}) + gain_T(t_n)(SSM(t_n) - SWI_T(t_{n-1})) \quad \text{(Eq. 4)}$$

The $gain_T$ factor at time *t* is derived using the formulation:

[4] https://dgg.geo.tuwien.ac.at

$$gain_T(t_n) = \frac{gain_T(t_{n-1})}{gain_T(t_{n-1}) + e^{-\frac{t_n - t_{n-1}}{T}}} \qquad \text{(Eq. 5)}$$

In Eq(4) and Eq(5), n refers to the latest, and $n-1$ to the previous *SSM* measurement.

Calculation of *SWI* based on recursive formulation can start, as new input *SSM* data are available from scatterometer measurements. However, in SWI processor, the SWI output product is provided in daily intervals. According to Eq(1) it can be shown that $SWI_T(t_c) = SWI_T(t_n)$, where t_c is the SWI calculation time and t_n is the time of last available SSM measurement.

Additionally, a quality indicator *qflag* is introduced. This quality flag is related to the number of available *SSM* measurements used for calculating SWI_T at time *t*. *qflag* is normalized by a factor of $\sum_{j=0}^{\infty} e^{-\frac{j}{T}}$ and expressed in percentage. So 100% indicates the ideal case and 0% shows that there were not enough *SSM* measurements available to calculate SWI. The recursive formulation of *qflag* is as follows:

$$qflag_T(t_i) = \begin{cases} 1 + qflag(t_{n-1}) \cdot e^{-\frac{t_n - t_{n-1}}{T}}, & \text{if SSM at } t_i \text{ is available } (t_i = t_n) \\ qflag(t_{n-1}) \cdot e^{-\frac{t_i - t_n}{T}}, & \text{if SSM at } t_i \text{ is unavailable} \end{cases} \qquad \text{(Eq. 6)}$$

Using a recursive form of the *SWI* (Eq. 4) gives the opportunity to implement the *SWI* estimation in near-real time processing environments, as no preceding *SSM* data are required. *SWI* is calculated using the current *SSM* data and the latest *SWI* information.

Calculation of *SWI* is started at any time with following initial values:

$SWI_T(t)=SSM(t)$, $gain_T(t)=1$, $qflag_T(t)=1$, $t_{n-1}=t$

3.2.3.2 Masking ASCAT SSM measurements before SWI calculation

Soil moisture measurements, which are affected by frozen or snow-covered surface, should be masked before using in SWI calculations.

Prior to product version 4.0, this masking utilized the SSF data derived from backscatter. In these previous versions, the masking of SSM is applied in two steps: At first, the SSF data are checked. If

SSF value for the individual DGG point is known then masking is done based on SSF information. Determination of SSF is succeeded through decision trees and by using information on normalized backscatter and freeze/thaw parameters. In the next step if SSF is unknown then the advisory quality flags provided in ASCAT input data are used for masking of SSM. These quality flags are some kind of climatological parameters, which are extracted from long-term snow and temperature observations and show the probability of frozen and snow-covered surface ranging between 0-100% with respect to time of year. Figure 3 and Figure 4 show two examples of snow and frozen quality flags in global scale. More information about ASCAT advisory flags can be found in Scipal et al. (2005).

In the new product version 4.0, masking is done using the advisory flags and frozen soil probabilities which accompany the SSM data of the new H29 ASCAT SSM input product. If the advisory flag is set to “frozen soil” then the respective SSM observation is masked. If the advisory flag is missing for a particular point, we fall back to the frozen soil probability and mask SSM observations with a probability over 90%. These probabilities are based on historic analysis of ERA5 data. Details on these advisory flags can be found in the ASCAT SSM H29 Product User Manual[5].

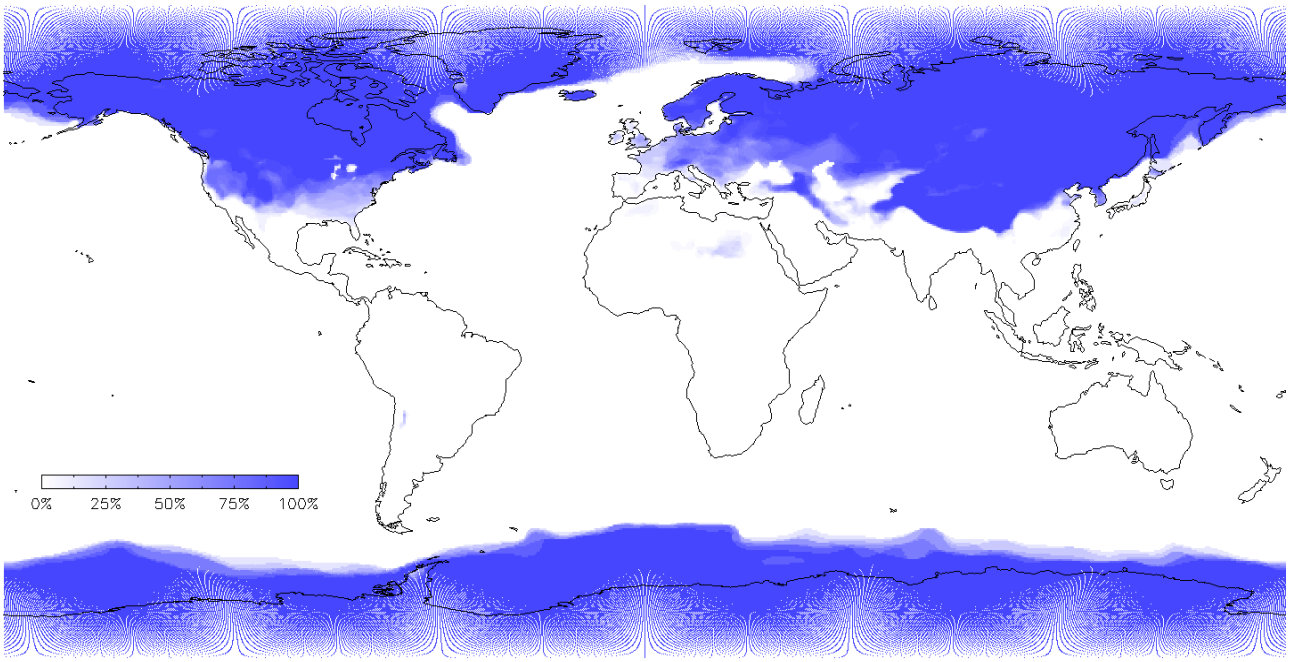


Figure 3: Frozen soil probability extracted from long-term ERA-40 modeled data on 1st of January

[5] https://hsaf.meteoam.it/Products/Detail?prod=H29

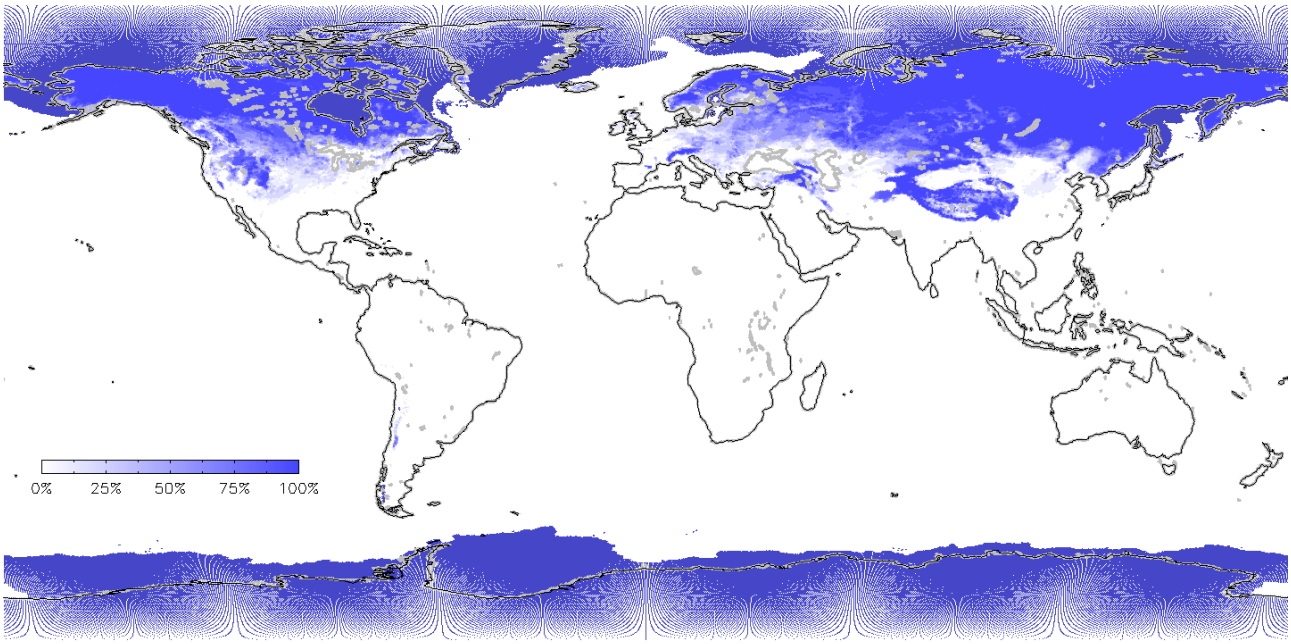


Figure 4: Snow cover probability extracted from long-term SSM/I data on 1st of January.

3.2.4 Example Output Data

Figure 5 shows a time series of SSM, SWI v3.2, SSF and QFLAG data for a point near Yakutsk in the Sakha Republic, Russia. On the left side it can be seen how the SSF changes from frozen to unfrozen as the air temperature climbs above 0°C, and how this is reversed in the autumn when temperatures become colder again. On the right side the SSM data is shown with points indicating the SSF state of the measurement. It can be seen that the SWI is kept constant when no valid SSM data is available and how it evolves during summer when SSM can be measured. The figure also shows how the QFLAG drops if no new valid data is available.

Figure 6 and Figure 7 show output images of SWI v3.2, SSF and QFLAG for March 11th 2010. It can be seen how data in northern areas is masked by the SSF and how this impacts the QFLAG. The striping in the QFLAG data in Figure 7 is due to the orbit geometry of the ASCAT sensor.

For masking the SWI in regions in which the soil moisture retrieval is difficult or not possible three layers of ancillary data are available. Their retrieval is explained below.

Topographic complexity

Soil moisture retrieval in mountainous regions can be difficult because of various error sources on the radar signal. Calibration errors due to the deviation from the reference ellipsoid influence of snow and ice cover as well as reduced sensitivity due to forest and rock are the main error sources.

The topographic complexity was calculated from GTOPO30 (USGS 1996) data which is a global digital elevation model with 30 arc seconds resolution (approx. 1km). For each grid point the standard deviation of elevation was calculated. The results were then normalized to values from 0 to 100 (Scipal et al. 2005).

Wetland fraction

The wetland fraction is based on the Global Lake and Wetlands Database (GLWD) (Lehner and Döll 2004). It gives the fraction of each grid cell that is covered by water in percent from 0 to 100.

Tropical forest mask

The tropical forest mask is based upon the fact that radar backscatter from C-Band is mostly stable over tropical rain forests (Birrer et al. 1982; Crapolicchio and Lecomte 2004). For ASCAT every grid point it is checked if the wet and dry references of the SSM algorithm are below 31.9dB and if the sensitivity (difference between highest and lowest backscatter observation) is below 1dB. It these thresholds are met it is assumed that the grid point is located over tropical rainforest.

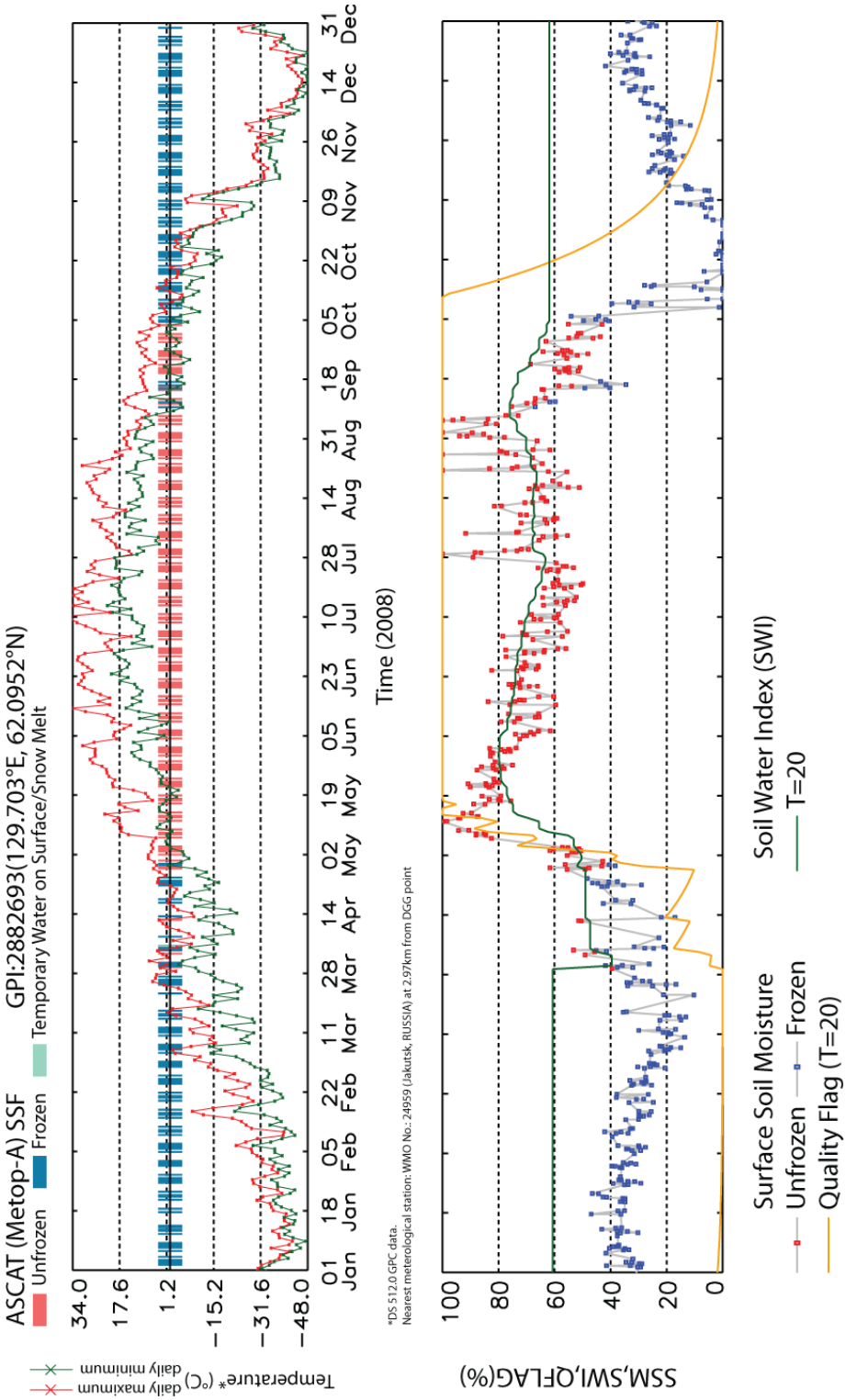


Figure 5: Left: Example illustration for SSF state plotted against daily minimum and maximum air temperature. Right: SSM input data and SWI v3.2 and QFLAG for T=20

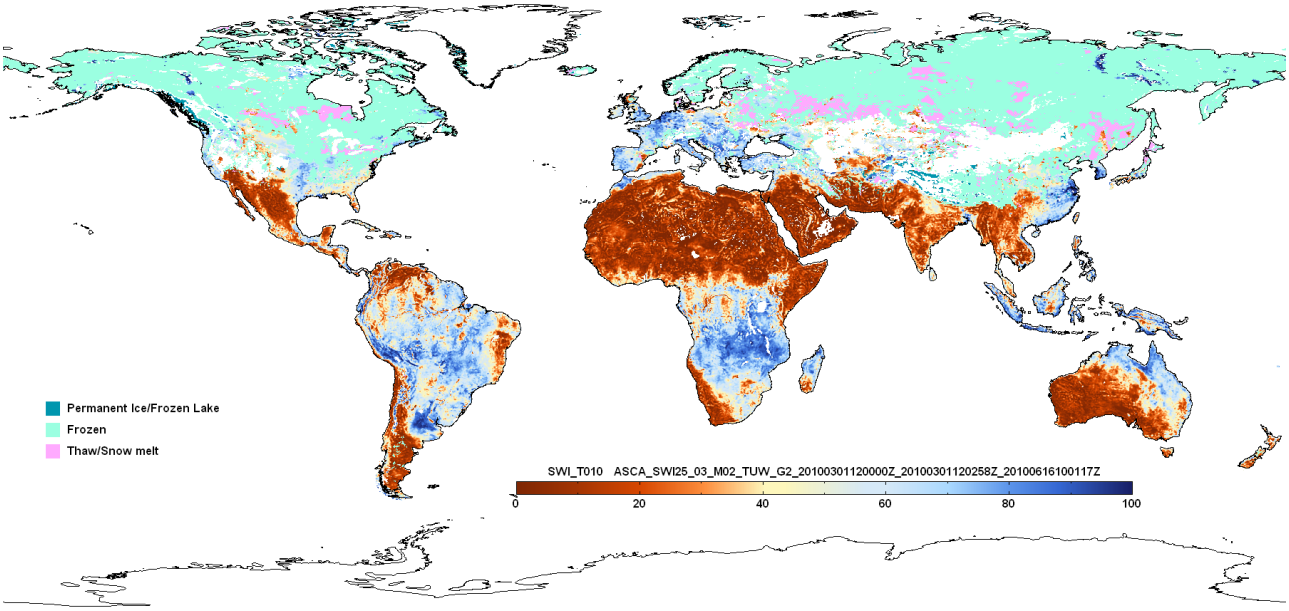


Figure 6: Output image of the SWI v3.2 calculated for T=10 overlaid by SSF for March 11th 2010.

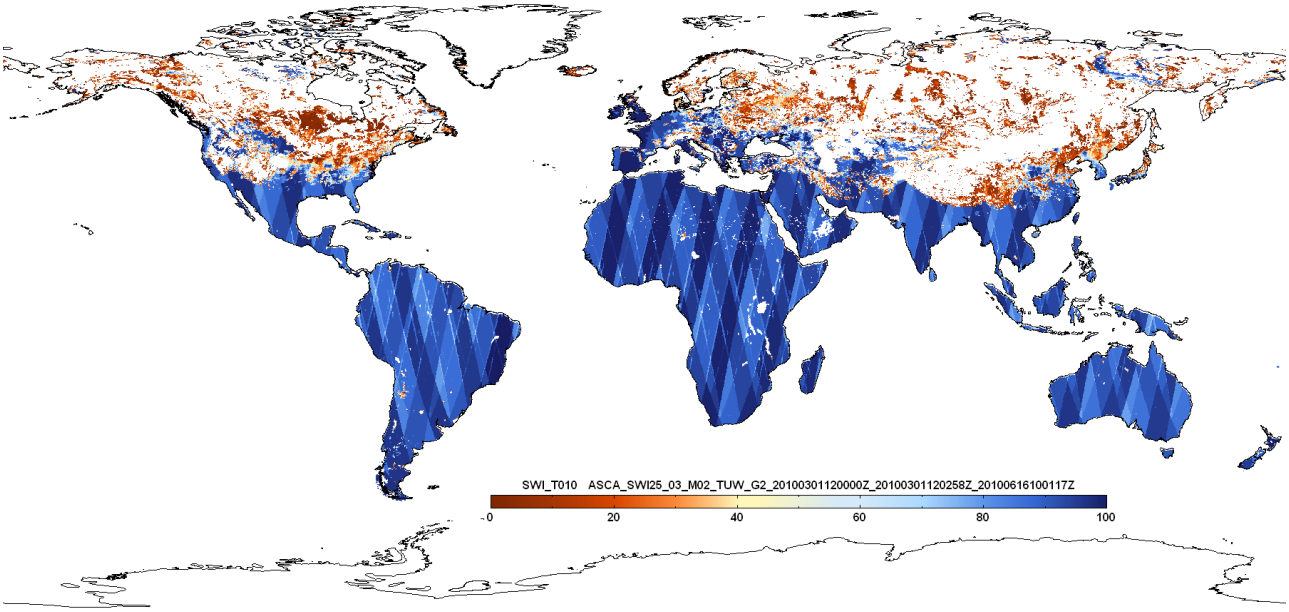


Figure 7: Output image of the quality flag calculated for T=10 for March 11th 2010

3.2.5 Limitations

The SWI algorithm cannot be applied in regions for which a surface soil moisture extraction from ASCAT C-band scatterometer data is not meaningful. This especially applies to high latitude and high altitude permafrost regions, glaciers, ice caps, dense forests (e.g. tropical rainforest) and deserts.

In areas with a high ground water table the SWI method does not work particularly for the deeper layers affected by the ground water.

The SWI and the SSF model both have shortcomings which are discussed in the respective sections 3.2.1.2 and 3.2.2.2.

Although SSF gives information on frozen and snow-covered surface, SSF is used only for masking SSM data in SWI calculations prior version 4.0.1. In version 4.0.1 the SSF flag is not used at all anymore and is only kept as a supporting dataset for the user and for backwards compatibility. In fact, the way of masking of SWI measurements are left to the users. SWI data can be masked by using quality flag, SSF data or one or more of the ancillary data sources which are delivered together with the SWI or by using some other external datasets.

3.3 The Product Accuracy

The product accuracy essentially relies on the accuracy of the input ASCAT Level2 surface soil moisture. An elaborated error analysis of this input has been performed before. It involves a Gaussian error propagation model, which tracks the different sources of errors and signal noise from radar backscatter to the final surface soil moisture product (Naeimi 2009).

Assessing the product accuracy of SWI V3.2 was carried out in close cooperation with external scientific users and partners of TU Wien using their site and data specific experiences (Brocca et al. 2011; Naeimi et al. 2013). The SWI data was compared to in-situ and simulated soil moisture data profile as regular quality monitoring exercise and the main results are reported in the Validation Report [GIOGL1_QAR_SWIV3-SWI10].

In product version 4.0, the input data-stream has been switched to the H29 ASCAT SSM product, and details on its accuracy can be found in its validation report [ASCAT_SSM_NRT_12_5km_H29_PVR_v0_1]. The quality of the V4.0 products are checked within the standard CLMS operational quality assessment cycle.

3.4 Risk of failure and Mitigation measures

Near-real time generation of SWI data essentially relies on the timely availability of ASCAT Level2 surface soil moisture data. If one of the sensors fails, SWI processing can continue without

interruption. Since less data would be available the quality flags of all T-values will show lower values because it takes more time to get new observations, especially in areas where the 2 or 3 instruments observe different regions per day

Should all sensors fail the SWI can no longer be calculated and the service would have to be put on hold until a suitable backup satellite is found.

The recursive reformulation of the SWI algorithm allows processing of ASCAT Level2 data to derive profile soil moisture at any time. In case of failure of the real-time data provision, also archived data from the EUMETSAT UMARF data archive can be used for alternative SWI calculation.

4 POST PROCESSING STEPS

The 2 products, SWI time series (SWI_TS) and 10-day averaged SWI (SWI10), are defined as independent post processing steps to the NRT SWI processing line. The input, in both cases, are the SWI products as netCDF4 images.

4.1 SWI10

The SWI10 product follows a very simple algorithm and is produced every 10 days. Ten days intervals are defined as follows: days 1 to 10, days 11 to 20, days 21 to end of month for each month of the year. SWI observations made during frozen surface conditions are masked before calculating the SWI10 product. Then, SWI10 is calculated by using only those measurements that show a SSF of "unfrozen" (Table 1). Because of this masking, the SSF is no longer present in the output SWI10 product since it would only contain unfrozen flags. Instead, a new layer is introduced containing the percentage of valid observations in the averaging period. The percentage is used since the length of the averaging period can vary for end of month products. The content of the SWI10 product is described in the Product User Manual.

The SWI10 processing is shared in the following steps:

1. Read SWI NRT data in HDF5 format of the last 10 days (check for completeness)
2. Mask invalid data in all images (Frozen SSF or NaN values)
3. Compute average for each grid point, while saving the number of valid observations in the averaging period.
4. Calculate percentage of valid observations in averaging period.
5. Save average values for the 8 SWI and 8 QFLAG layers as well as percentage of valid observations.

4.2 SWI_TS

The SWI_TS product does not change the contents of the SWI product but changes the format to make them easily available for time series analysis. SWI_TS is currently available for the SWI Archive and not in NRT to guarantee a consistent time series of the best possible quality.

The SWI images are saved on a regular 0.1° grid which leads to oversampling in high latitude regions. For the time series, the data are only given on the points of the original grid (DGG) used for the ASCAT data at TU Wien. The DGG has a grid point spacing of 12.5 km containing a total of 3 264 391 points of which 839 826 are over land areas. This grid is oversampled by the regular 0.1° image

which contains a total of 6 480 000 points of which are approx. 1 814 400 over land. This means that significantly fewer time series files have to be saved using the DGG compared to the regular 0.1° grid without losing any information.

The algorithm to change the format from daily images to time series is the following:

1. Read SWI HDF5 images in bulk (100-500 images at once), depending on the working memory of the machine
2. Perform reverse lookup between regular 0.1° grid and DGG land points.
3. Stack the images into 3D array structure.
4. Read time series of each grid point out of 3D array.
5. Save/Append time series to time series files.

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